

# **SE4020 - Mobile Application Development & Design - Assignment 2**

Part B: VisionOS

## AI Prompt Usage Documentation

**Submitted by:**

## AI Usage Log

This log records how AI tools were used to build NurseryConnectVision, with what was accepted, rewritten, or rejected. The goal is honesty and specificity; the marks reward showing *judgement* over the AI's output, not just that a tool was used.

### Primary

### tool:

Claude Code (Anthropic) runs in the terminal against the project repo.

**Why this tool:** it can read the actual project files, build/test against the visionOS simulator, and iterate on compile errors in-loop, so its output is grounded in the real codebase rather than generic snippets. visionOS / RealityKit APIs change fast, so being able to *build and verify* each step mattered more than raw generation.

Note: prompts below are reconstructed/representative of the working session. Where a prompt was short and conversational (e.g. "Proceed with D"), it stands for "implement the next phase per the phase doc." Replace any bracketed placeholders with the exact text if your rubric wants verbatim.

## Entry 1 - Phase planning

- **Prompt:** "check the docs/phases and start from each phase" (after a planning pass that splits the build into Phases A–G with rubric mappings).
- **Response:** a 7-phase plan (data layer → roster window → volume → spatial interactions → polish/accessibility → testing → submission docs), each phase doc listing files, code, and acceptance criteria.
- **Accepted / rewrote / rejected:** accepted the phase structure; it matched the rubric weighting (25 + 20 functionality/UX, 15 reflections, etc.). Enforced a manual-review gate myself implement one phase, stop, I review and commit before the next.

## Entry 2 - Phase A: data layer + tests

- **Prompt:** implement Phase A (models, store, view model, unit tests).
- **Response:** Child (reused), DiaryEntry + WellbeingLevel, IncidentReport, NurseryStore (@Observable, seeded (referenceDate:) with 5 children), DashboardViewModel, and 5 Swift Testing tests.

- **Accepted / rewrote / rejected:** accepted. Verified the seeded data made sense (Noah carries the one incident) and that incidents the property and incidents (for:) the method coexist.

## Entry 3 - Nursery volume (programmatic RealityKit)

- **Prompt:** implement Phase C (build the room in code; volumetric window).
- **Response:** ChildTagComponent, NurseryRoomBuilder (floor + walls + pegs + name plates), NurseryRoomView (RealityView), second volumetric WindowGroup, ornament toggle → open/close.
- **Accepted / rewrote / rejected:** accepted, after fixing three real compile issues the AI and I resolved in-loop:
  - Cannot find type 'UUID' → added import Foundation to ChildTagComponent.swift.
  - registerComponent() unavailable → added import RealityKit to the App file.
  - 'Scene' is ambiguous (RealityKit also defines Scene) → qualified as some SwiftUI.Scene.
  - Also diagnosed a *stale-IDE* error (red "Cannot find NurseryRoomBuilder" while the CLI build passed) as a DerivedData/Previews artifact, fixed by clean + DerivedData wipe - **no code change**. Good example of not "fixing" code that wasn't broken.

## Entry 4 - The three spatial interactions

- **Prompt:** "Proceed with D" implement tap→anchored panel, drag-to-rotate, spatial chime.
- **Response:** ChildDayPanel, AudioCue (spatial chime.wav, no-op if absent), and the 3-closure RealityView(content/update/attachments) wiring with SpatialTapGesture + DragGesture.
- **Accepted / rewrote / rejected:** accepted the logic. **Rewrote the panel placement** after testing in the simulator: the panel originally anchored *beside* the peg at label height and overlapped the floating name plates I moved it to float *above and in front* of the selected peg (+[0, 0.22, 0.16]) and made it track the peg's X so it can't run off the edge.

## Entry 6 - Polish + Accessibility

- **Prompt:** "update the UI accordingly then proceed with E" (after reviewing a simulator screenshot showing the panel/label overlap).
- **Response:** centred name plates (measure text mesh width), taller walls, DirectionalLight
  - shadow, styled HoverEffectComponent, per-peg AccessibilityComponent, and Reduce Motion / Reduce Transparency handling.

- **Accepted / rewrote / rejected:** accepted. The screenshot-driven feedback loop was the difference-maker the fixes came from *seeing* the rendered volume, not from the spec alone.